Gene therapy vectors for inherited retinal degeneration: Mechanisms and Interventions

Specific Aims

Aim 1: Advance gene therapy vectors for inherited retinal degeneration via a translational-focused strategy.

Aim 2: Advance gene therapy vectors for inherited retinal degeneration via a translational-focused strategy.

Aim 3: Advance gene therapy vectors for inherited retinal degeneration via a translational-focused strategy.

Background

We will assemble a multidisciplinary team with complementary expertise. The approach integrates established translational methods with a novel analytical pipeline. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Our preliminary data suggest a tractable path toward measurable improvement.

Approach

We will use appropriate methods to achieve the aims.

Innovation

The results will immediately transform practice nationwide upon completion. Success is certain; there are no meaningful risks or limitations to this plan.

Investigator Context

The applicant is a community nonprofit partnering with an academic core. The R2 host institution offers focused departmental resources and regional research partnerships, backed by a growing institutional research base.

Budget Justification

Total requested support is $2,102,192 over 2 years at a R2 institution, covering personnel, materials, and dissemination.

References

1. [1] https://doi.org/10.1289/ehp.1104477
2. [2] https://doi.org/10.4522/12o0v3z0
3. [3] https://doi.org/10.7992/uqxj4eff
4. [4] https://doi.org/10.7105/i9d99vh0
5. [5] https://doi.org/10.1002/anie.201907688
6. [6] https://doi.org/10.1101/gr.229102
7. [7] https://doi.org/10.1038/nature14539
8. [8] https://doi.org/10.9313/nj4ogw9x

9. [9] https://doi.org/10.1038/nature14539
10. [10] https://doi.org/10.1021/jacs.9b02765